In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [12]:
```
df = pd.read_csv(r"R:\rana\rana\train.csv")
```

In [20]:
```
print('Database shape:',df.shape)
print('head:',df.head())
print('info:',df.info())
```

```
Database shape: (891, 12)
head:    PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
info: None
```

In [21]:
```
print(df.describe())
```

```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

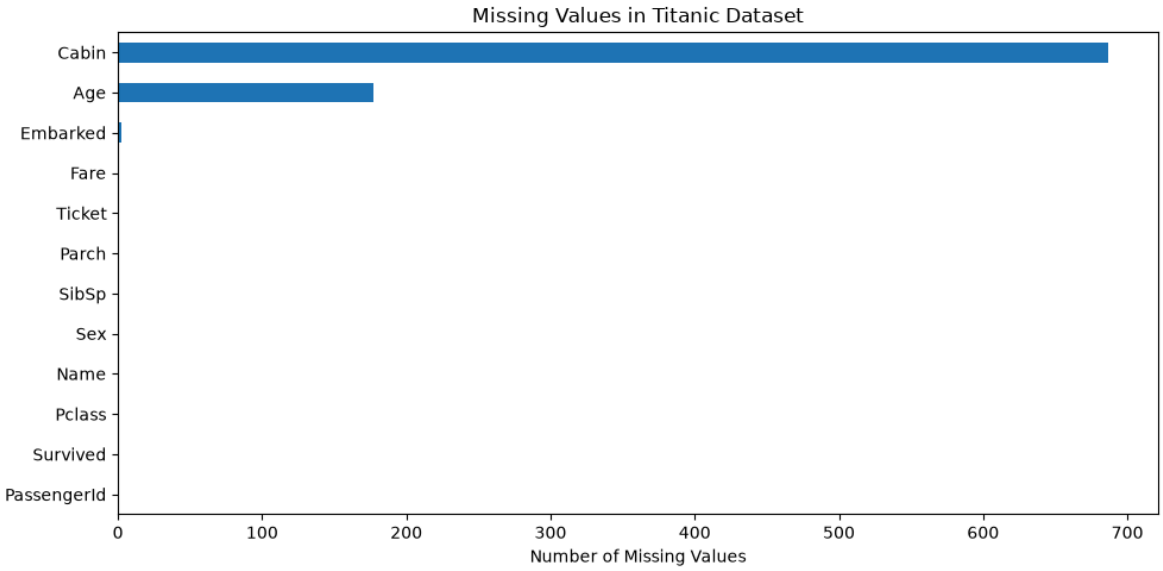
In [22]:
```
print(df.isnull().sum())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [23]:
```
plt.figure(figsize=(10, 5))

df.isnull().sum().sort_values().plot(kind="barh")

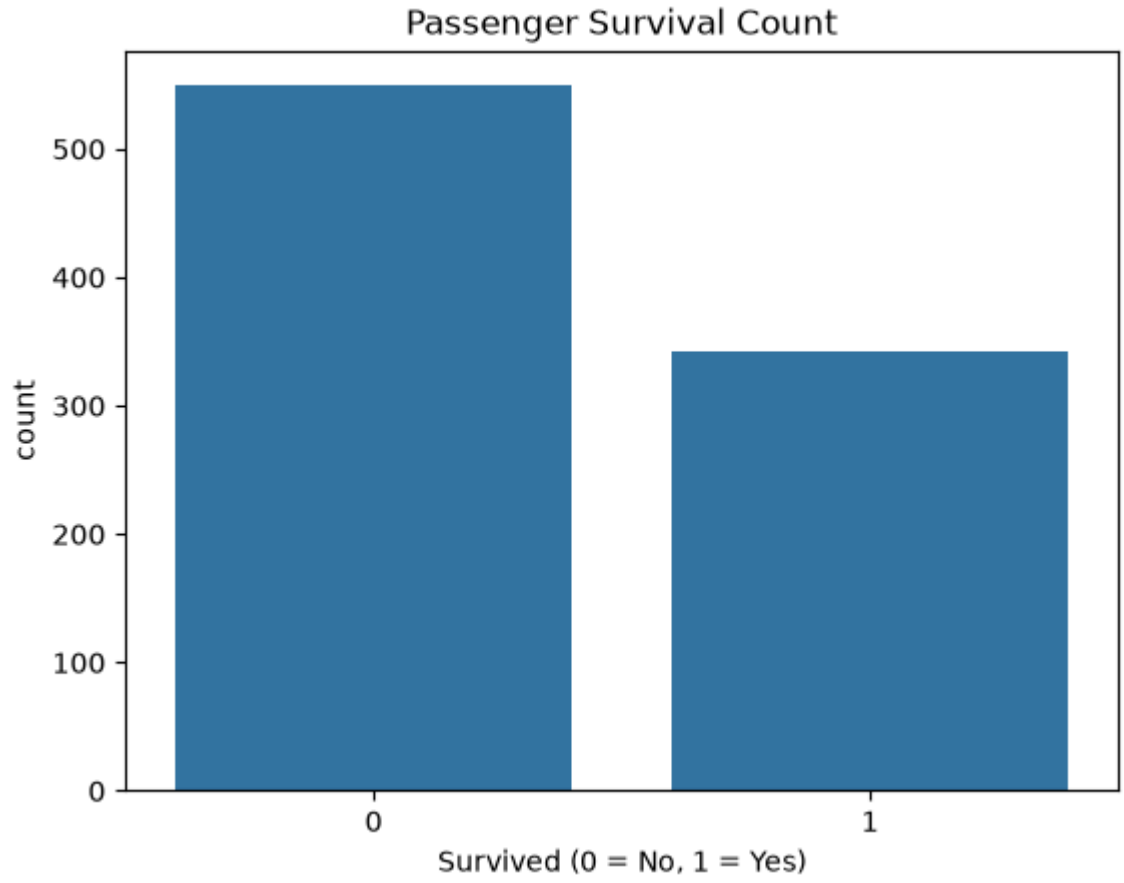
plt.title("Missing Values in Titanic Dataset")
plt.xlabel("Number of Missing Values")
plt.tight_layout()
plt.show()
```



In [24]:
```
print(df["Survived"].value_counts())

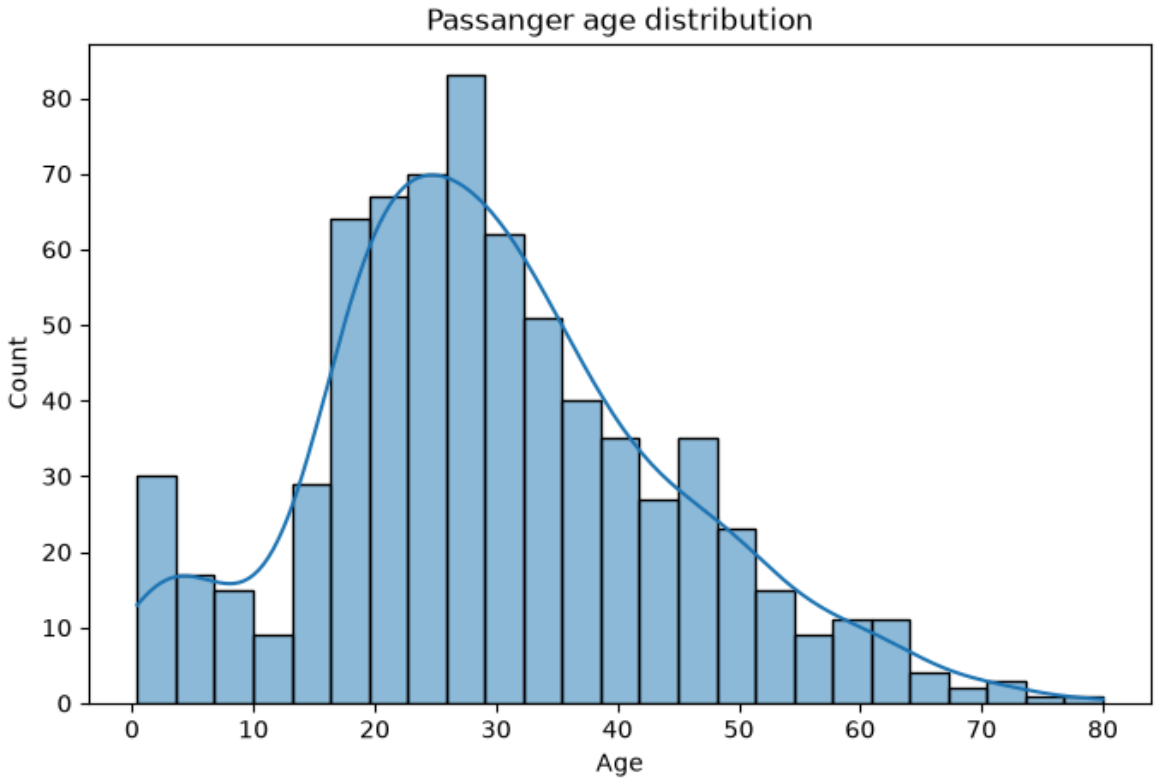
sns.countplot(data=df, x="Survived")
plt.title("Passenger Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.show()
```

```
Survived
0    549
1    342
Name: count, dtype: int64
```



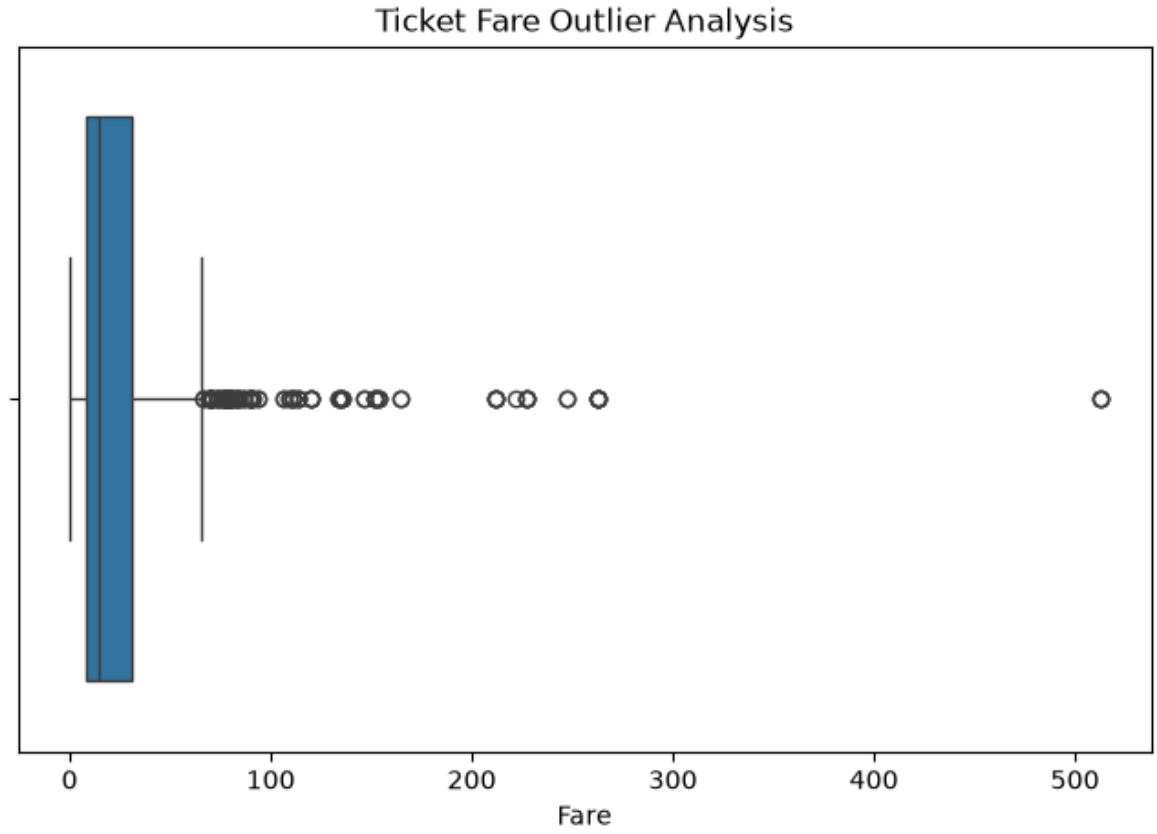
In [27]:
```
plt.figure(figsize=(8,5))
sns.histplot(data=df, x="Age", bins=25, kde=True)
plt.title('Passanger age distribution')
```

```
plt.xlabel('Age')
plt.show()
```



In [28]:
```
plt.figure(figsize=(8,5))
sns.boxplot(data=df, x='Fare')

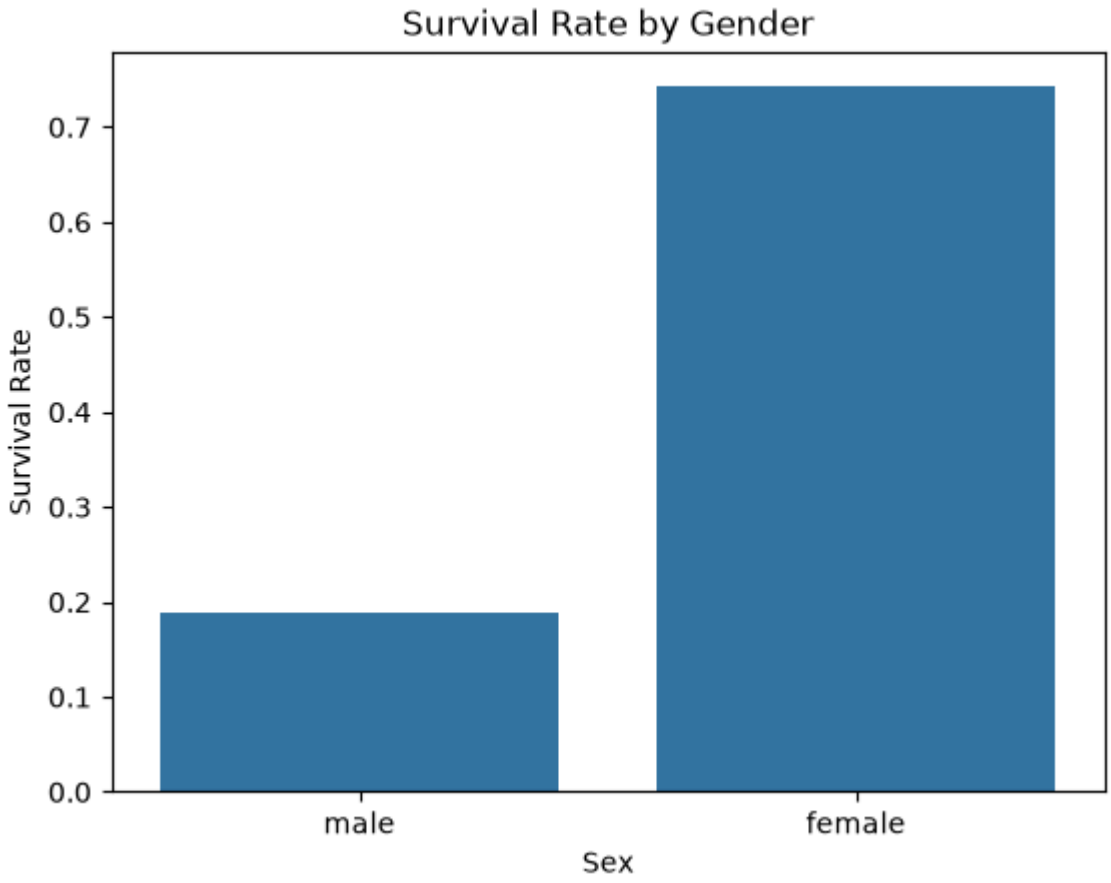
plt.title("Ticket Fare Outlier Analysis")
plt.show()
```



In [29]:
```
sns.barplot(data=df, x="Sex", y="Survived", errorbar=None)

plt.title("Survival Rate by Gender")
plt.ylabel("Survival Rate")
plt.show()

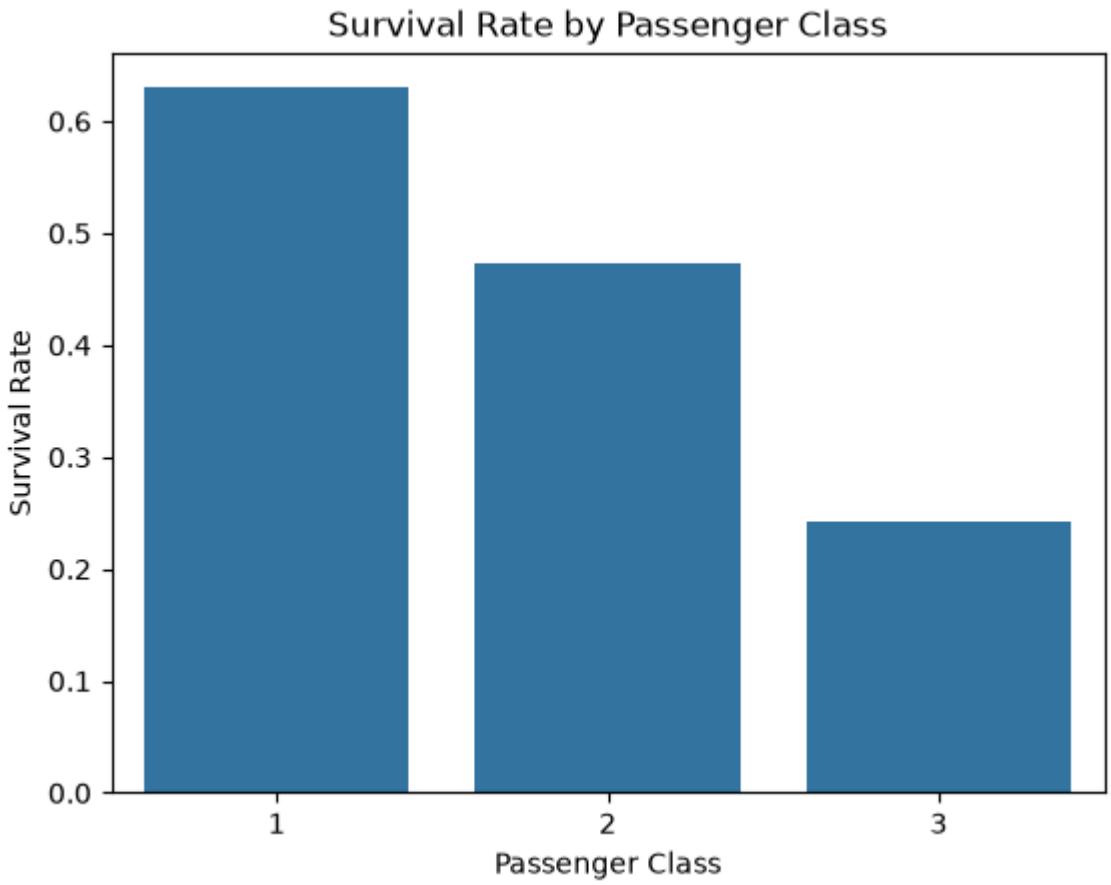
print(df.groupby("Sex")["Survived"].mean() * 100)
```




```
Sex
female    74.203822
male      18.890815
Name: Survived, dtype: float64
```

In [30]:
```
sns.barplot(data=df, x="Pclass", y="Survived", errorbar=None)

plt.title("Survival Rate by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Survival Rate")
plt.show()
```

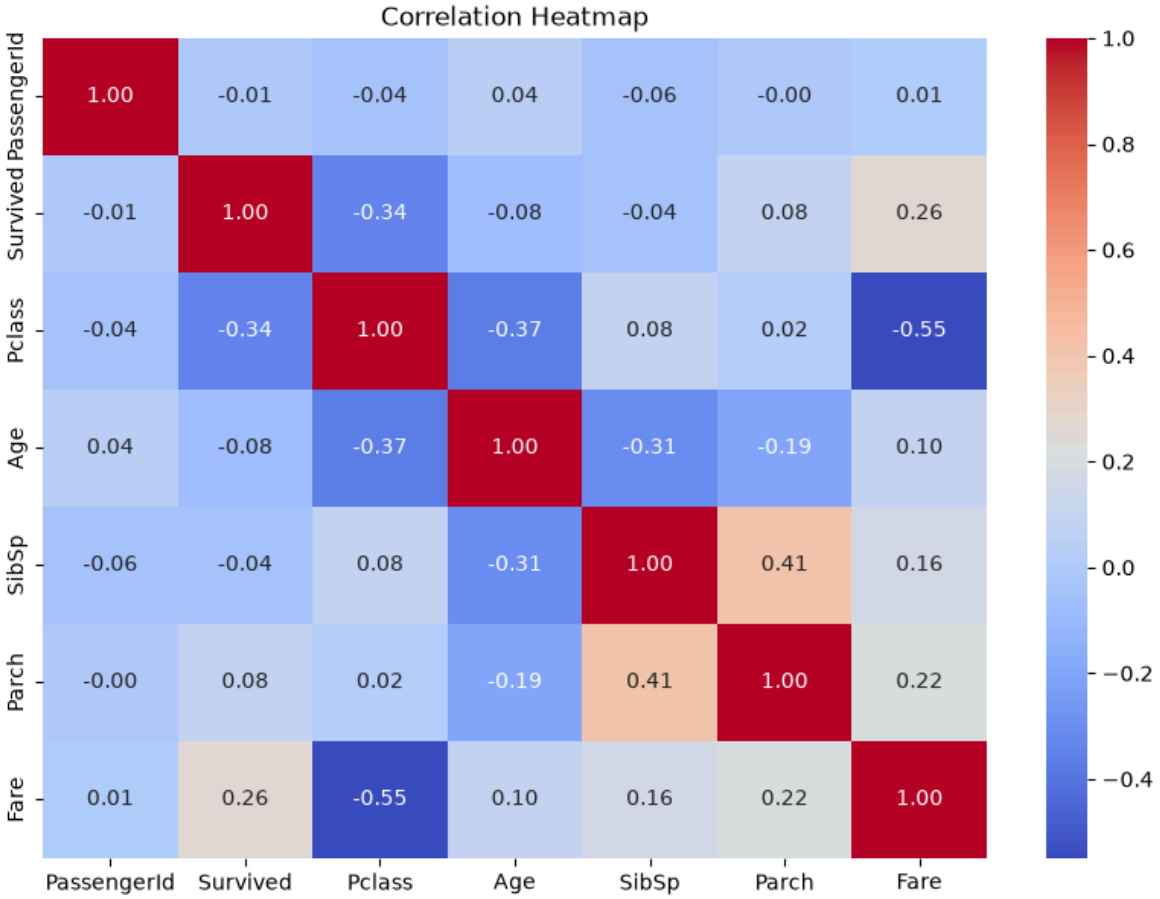


In [31]:
```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(10, 7))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

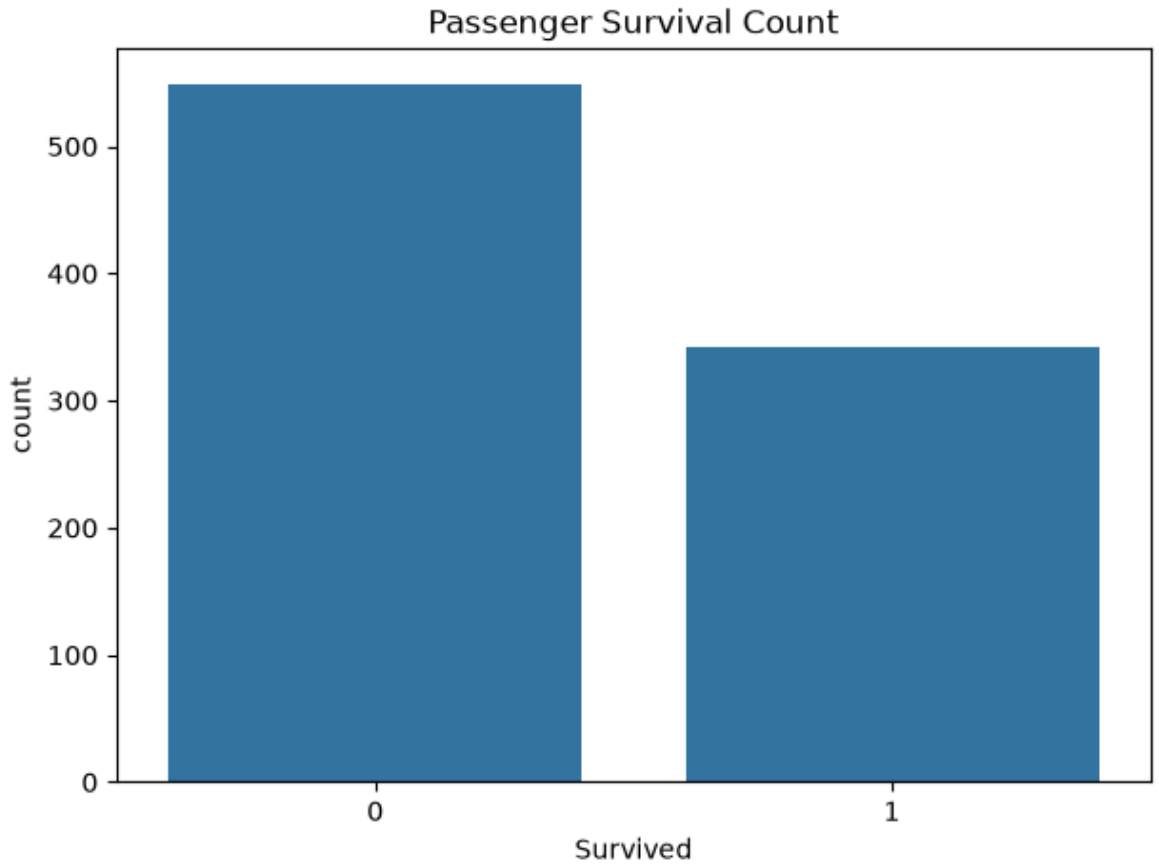
plt.title("Correlation Heatmap")
plt.show()
```



In [32]:
```
import os

os.makedirs("visualizations", exist_ok=True)

plt.figure(figsize=(7, 5))
sns.countplot(data=df, x="Survived")
plt.title("Passenger Survival Count")
plt.savefig("visualizations/survival_count.png",
            dpi=150, bbox_inches="tight")
plt.show()
```



In [33]:
```
print("TOTAL PASSENGERS:", len(df))
print("TOTAL SURVIVORS:", df["Survived"].sum())

print("SURVIVAL RATE:",
      round(df["Survived"].mean() * 100, 2), "%")

print("\nMissing values:")
print(df.isnull().sum())

print("\nEDA completed successfully!")
```

```
TOTAL PASSENGERS: 891
TOTAL SURVIVORS: 342
SURVIVAL RATE: 38.38 %

Missing values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64

EDA completed successfully!
```

In [ ]: